UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2007 question paper

5090 BIOLOGY

5090/03 Paper 3 (Practical Test), maximum raw mark 40

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International Examinations

1 (a) (i) 1 At least 5 cm, clear and realistic [D.3]
2 Grain shown + root(s)
3 Coleoptile/cotyledon observed.

Labels: seed/grain/testa ;
one from root/radicle/coleoptile/cotyledon [2]

(ii) Both measurements, (both with 1d.p. if cm.) units once ;
R: if no indication where measured
Expression and calculation ;
Mag. Well expressed ; [up to 2 d.p. and 0.2 rounding] [3]

(b) (i) test for starch: brown or other result ;
Test for r/s: blue/no colour change ; [2]

(ii) add Benedict's ;
equal amount etc. ;
Heat – w.bath stated or implied ; [3]

(c) test for starch: brown etc. ;
test for r/s: green/yellow/orange/brown qualified ; [2]
A: up to 1 each for conclusions in these 3 sections of (b) and (c).

(d) enzyme in seed/diastase ;
changes/digests starch ;
to (reducing) sugar/maltose/glucose ;
which diffuses out/through membrane/revealed in test ; [up to 3]
Ignore ref osmosis

(e) grow seedlings of maize and rice ;
method of preparation (e.g. chop) ;
tests as before ;
same temp. ;
equal amounts of reagents/material ;
equal amount of starch ;
same time scale ; OVP ; [up to 4]

[Total: 22]

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2 (a) (i) two measurements both with units ; *[1st d.p if cm]*
working including division by 2 ;
answer correct in cm^2 (mm^2) ; [3]

(ii) Neatly traced outline ;
obvious tally system used ;
allowance for incomplete squares ;
realistic result clearly recorded, with units ; [4]

(iii) Advantage: speed ; *R. easy/simple*
fieldwork ;
only ruler needed ; OVP ; [up to 1]
Disadvantage: degree of inaccuracy ;
shape of leaf ; OVP ; [up to 1]

(b) (i) thickness/height ; [1]

(ii) short diffusion pathway ;
for CO_2 especially ;
light catching ;
for p/s (energy) ; [up to 2]

(c) (i) green ;
flat ;
veined ;
large s/a ; [up to 3]

(ii)

W1		W2
dicot.	v	monocot ;
net veined	v	parallel veins ;
leaf stalk	v	sessile ;
leaf only	v	node present ;
two sides differ	v	both sides alike ;
broad	v	narrow ;
jagged	v	smooth ;

stated colour difference ;
stated difference in hair distribution ; [up to 3]

[Total : 18]

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